

Transistors

- ❖ Transistor was invented by **John Bardeen**, Walter Brattain, and William Shockley in 1947 (got Nobel prize Bardeen was physicist).
- ❖ One of the most important invention in the area of science and technology.
- ❖ A **transistor** is a three-terminal semiconductor device that controls the flow of current.
- ❖ Transistor word has come from the two words TRANSFER-RESISTOR, low input resistance to high output resistance and vice versa.

Types of Transistor

- **Bipolar Junction Transistor (BJT)**
 - NPN
 - PNP
- **Field Effect Transistor (FET)**
 - JFET(Junction Field Effect Transistor)
 - MOSFET(Metal oxides semiconductor Field Effect Transistor)

What Are Bipolar Junction Transistors?

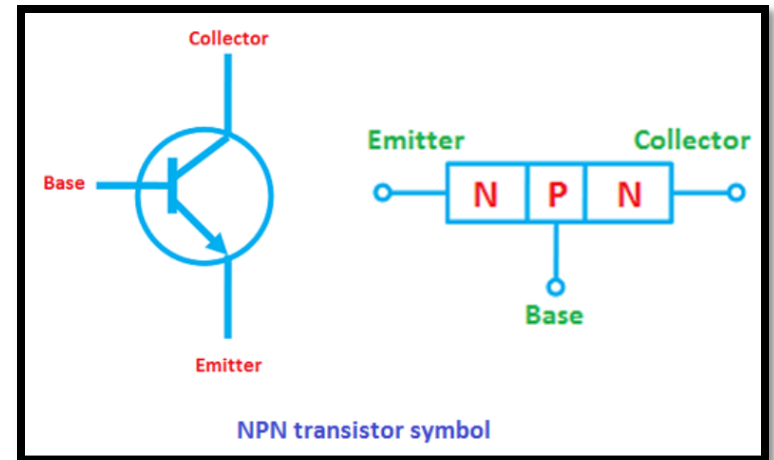
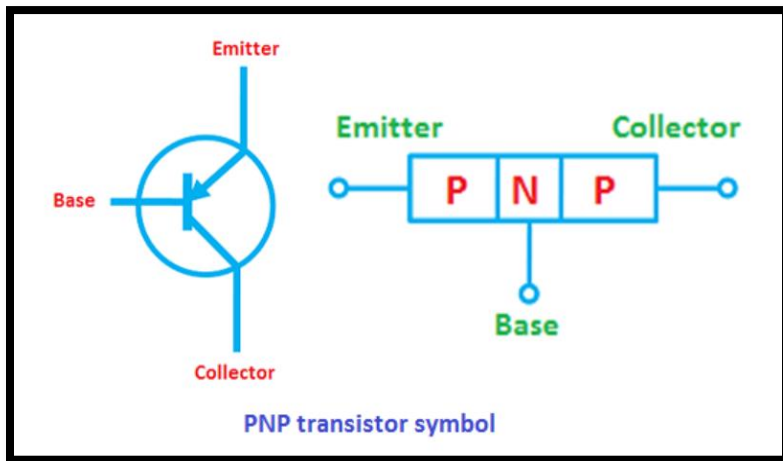
- ❖ Unlike semiconductor diodes which are made up from two pieces of semiconductor material (both P- and N-) to form one simple p-n junction. The bipolar transistor uses one more layer of semiconductor material to produce a device with properties and characteristics of an amplifier.
- ❖ If we join together two individual single diodes back-to-back, this will give us two p-n junctions connected together in series which would share a common *Positive* (P) or *Negative* (N) terminal.
- ❖ The fusion of these two diodes together produces a three layer, two junction, three terminal device forming the basis of a *Bipolar Junction Transistor* or BJT for short.
- ❖ Bipolar junction transistors are three terminal active devices made from different semiconductor materials that can act as either an insulator (fully-off) or a conductor (fully-on) by the application of a small signal voltage.

- The transistor's ability to change between these two states enables it to have two basic functions: "Switching" (digital electronics) or "Amplification" (analogue electronics). **Then bipolar transistors have the ability to operate within three different regions:**
- Active Region – the transistor operates as an amplifier with the collector current, I_c equal to: $\beta \cdot I_B$
- Saturation – the transistor is "Fully-ON" operating as a switch with the collector current, I_c is equal to: $I(\text{saturation})$
- Cut-off – the transistor is "Fully-OFF" operating as a switch with the collector current, I_c equal to: 0
- There are two basic types of bipolar transistor construction, **PNP** (Positive-Negative-Positive) and **NPN** (Negative-Positive-Negative). This notation basically describes the physical arrangement of the P-type and N-type semiconductor materials from which they are made.
- The **Bipolar Transistor** basic construction consists of two p-n junctions producing three connecting terminals with each terminal being given a name to identify it from the other two. These three terminals are known and labelled as the *Emitter* (E), the *Base* (B) and the *Collector* (C) respectively.

Constructional details of BJT

A transistor has **three Region** and hence **three terminals**:

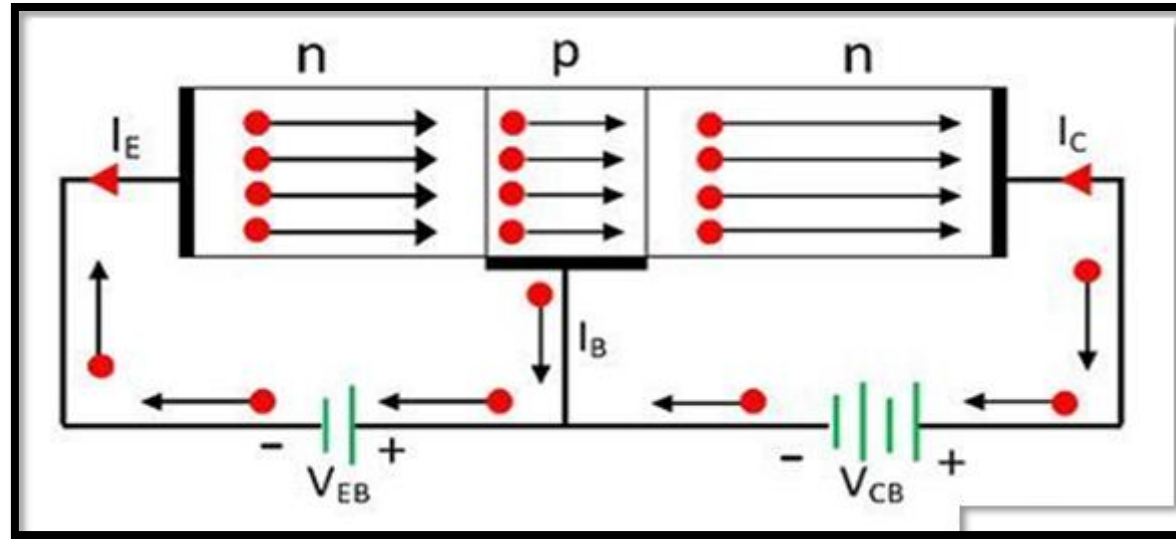
1. **Emitter (E)** – emits charge carriers and it is heavily doped
2. **Base (B)** – controls the current ,lightly doped and smaller size
3. **Collector (C)** – collects charge carriers either lightly or moderately doped and larger in size as compare to emitter



Arrow is always on the emitter and it indicates the direction of current for normal biasing of the transistors.

Normally emitter base junction is forward biased and collector based junction is reverse biased .

Working of the transistor



- ❖ Electrons are repelled from emitter and enter into the base region this causes emitter current which flows from the **emitter** into the **base**
- ❖ Since the base is thin, a small number of electrons recombine in the base and the loss of electron is recovered by the supply of electrons from collector base bias voltage → **base current (I_B)**
- ❖ Remaining **most of the electrons pass through it** and are attracted by the collector → **collector current (I_C)**
- ❖ Hence: **$I_E = I_B + I_C$**
- ❖ For efficient working of transistor, collector current should be as large as possible component of emitter current.

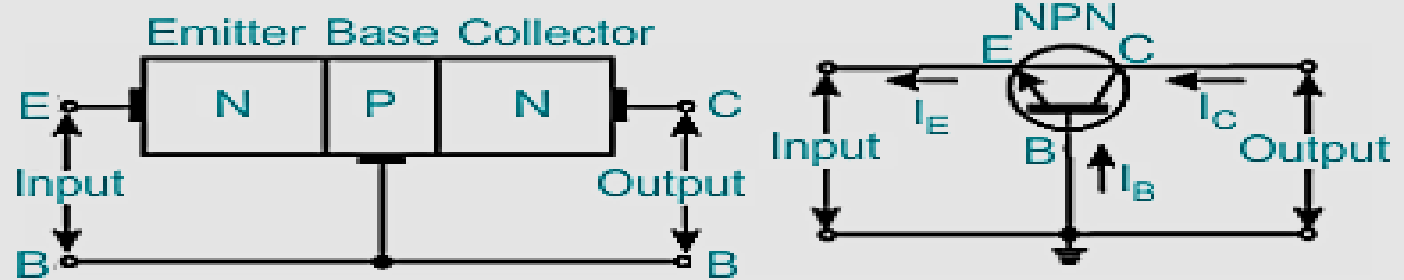
Bipolar Transistor Configurations

- ❖ The bipolar transistor is a three terminal device, there are basically three possible ways to connect it within an electronic circuit with one terminal being common to both the input and output signals.
- ❖ Each method of connection responding differently to its input signal within a circuit as the static characteristics of the transistor vary with each circuit arrangement.

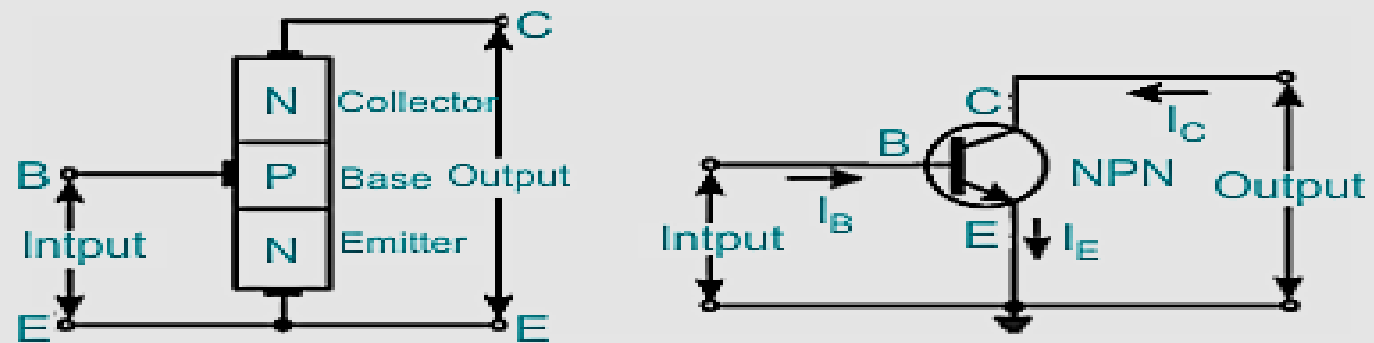
Junction	Active Region (Amplifier)	Saturation Region (Closed Switch)	Cut-off region (Open Switch)
Emitter-Base (EB)	Forward Biased	Forward Biased	Reverse Biased
Collector Base (CB)	Reverse Biased	Forward Biased	Reverse Biased

- ❖ The **biasing of the internal junctions** (EB and CB) defines the operating region, not the external configuration (CE, CB, CC), though the voltage polarities relative to the common terminal shift in different configurations

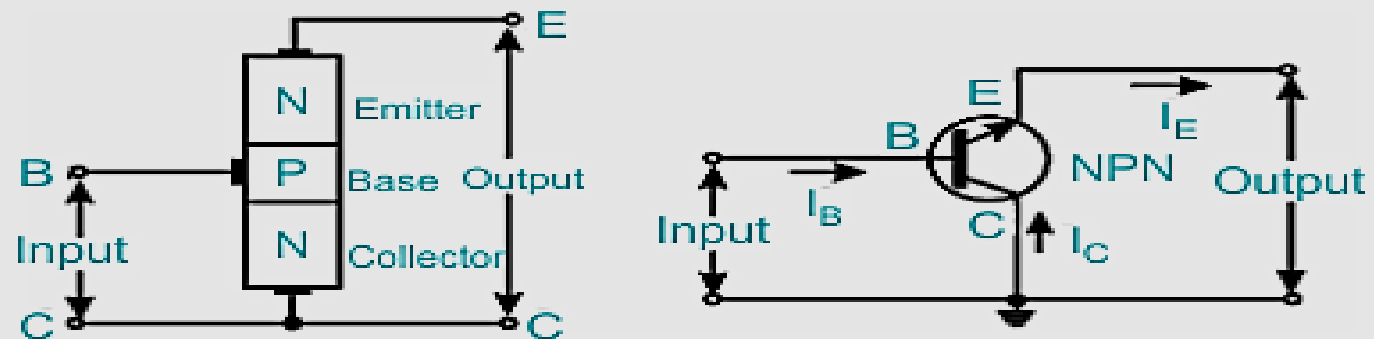
Active Region	CB	CE	CC
EB junction	Forward bias	Forward bias	Forward bias
CB junction	Reverse Bias	Reverse Bias	Reverse Bias
Connection	Base is common	Emitter is common	Collector is common
Input	Emitter Emitter current I_E , Emitter-Base voltage V_{EB}	Base Base current I_B , Base-Emitter voltage V_{BE}	Base Base current I_B , Base-Collector voltage V_{BC}
Output	Collector Collector current I_C , Collector-Base voltage V_{CB}	Collector Collector current I_C , Collector-Emitter voltage V_{CE}	Emitter Emitter current I_E , Emitter-Collector voltage V_{EC}
Characteristics	High voltage gain, low input impedance, no current gain	High voltage and current gain	High input impedance, low output impedance, unity voltage gain
Application	Used in RF/high-frequency applications	Most widely used for amplification, switching	Used for buffer/impedance matching



Common Base Configuration



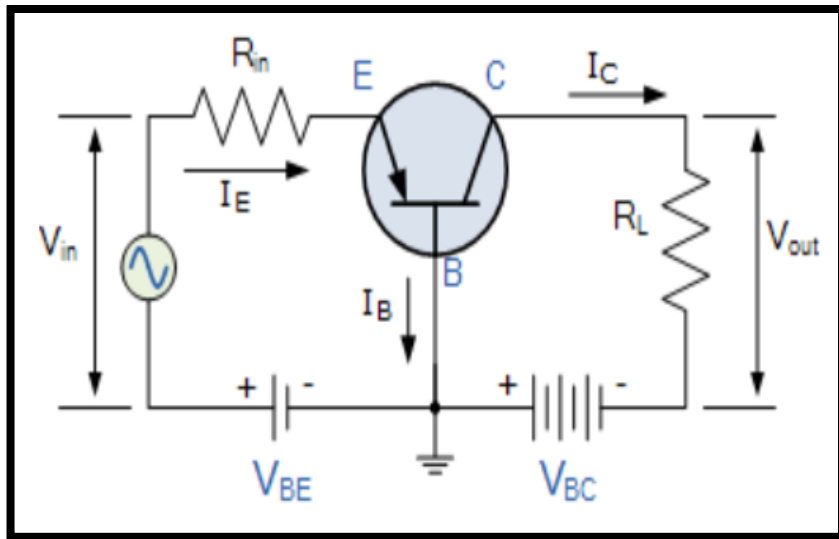
Common Emitter Configuration



Common Collector Configuration

Common Base (CB) Configuration

- In the **Common Base** or “grounded base configuration”, the transistors BASE terminal is common to both the input signal and the output signal. The input signal is applied between the transistors base and its emitter terminals.
- The corresponding output signal is taken from between the base and the collector terminals. The base terminal can be either grounded or can be connected to some fixed reference voltage point.



- The input current flowing into the emitter is quite large as its the sum of both the base current and collector current respectively thus resulting in a current gain that is unity or less for this circuit.
- Thus, the common base configuration “attenuates” the input signal.

- ❑ This type of Common Base Amplifier configuration is a non-inverting voltage amplifier circuit, in that the signal voltages V_{in} and V_{out} are “in-phase”.
- ❑ This type of transistor arrangement is not very common due to its unusually high voltage gain characteristics. Its input characteristics represent that of a forward biased diode while the output characteristics represent that of an illuminated reverse bias diode.
- ❑ This type of bipolar transistor configuration has a high ratio of output to input resistance giving it a value of “Resistance Gain”.
- ❑ The common base circuit is generally used in single stage amplifier circuits such as microphone pre-amplifier or radio frequency (R_f) amplifiers due to its very good high frequency response.

❑ **Current Amplification Factor /Current gain (α):**

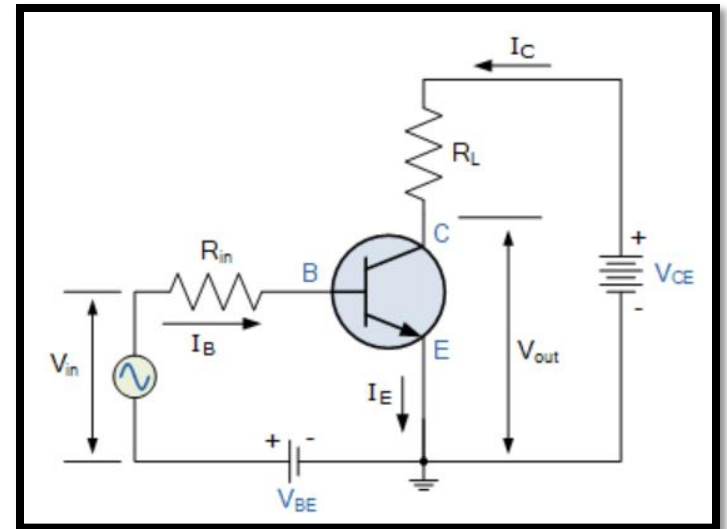
The current amplification factor is defined as the ratio of changes in Collector current (I_C) to the change in emitter current (I_E).

$$\alpha = I_C / I_E \text{ (at constant } V_{CB})$$

The value of α is always less than unity. The practical value of transistors lie between 0.95 and 0.99.

Common Emitter (CE) Configuration

- ❑ In the **Common Emitter** or grounded emitter configuration, the input signal is applied between the base and the emitter, while the output is taken from between the collector and the emitter.
- ❑ This type of configuration is the most commonly used circuit for transistor based amplifiers and which represents the “normal” method of bipolar transistor connection.
- ❑ The Common Emitter Amplifier configuration produces the highest current and power gain of all the three bipolar transistor configurations. This is mainly because the input impedance is LOW as it is connected to a forward biased PN-junction, while the output impedance is HIGH as it is taken from a reverse biased PN-junction.
- ❑ The emitter current for a common emitter configuration is defined as: $I_E = I_C + I_B$.



- ❑ As the load resistance (R_L) is connected in series with the collector, the current gain of the common emitter transistor configuration is quite large.
- ❑ Current gain $\beta = I_C/I_B$. Typically, Beta has a value between 20 and 200 for most general purpose transistors.
- ❑ The electrical relationship between these three currents I_E , I_C and I_B is determined by the physical construction of the transistor itself.
- ❑ Thus any small change in the base current (I_B) will result in a much larger change in the collector current (I_C).
- ❑ This configuration has a greater input impedance, current and power gain than that of the CB configuration but its voltage gain is much lower.
- ❑ The common emitter configuration is an inverting amplifier circuit.
- ❑ This means that the resulting output signal has a 180° phase-shift with regards to the input voltage signal

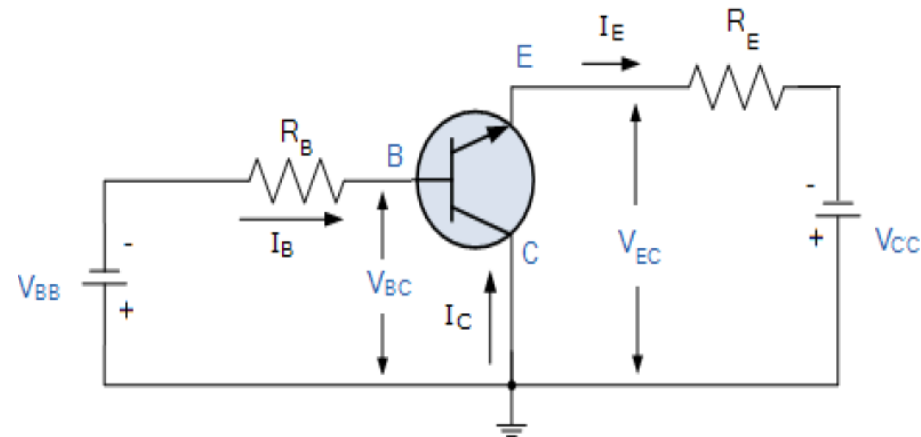
$$\text{Alpha, } (\alpha) = \frac{I_C}{I_E} \quad \text{and} \quad \text{Beta, } (\beta) = \frac{I_C}{I_B}$$

$$\therefore I_C = \alpha \cdot I_E = \beta \cdot I_B$$

$$\text{as: } \alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{1 - \alpha}$$

Common Collector (CC) Configuration

- ❑ In the **Common Collector** configuration, the collector is connected to ground through the supply, thus the collector terminal is common to both the input and the output.
- ❑ The input signal is connected directly to the base terminal, while the output signal is taken from across the emitter load resistor as shown. This type of configuration is also known as a **Voltage Follower** or **Emitter Follower** circuit.
- ❑ The common collector or emitter follower configuration is very useful for impedance matching applications because of its very high input impedance.
- ❑ The load resistance of the CC transistor receives both the base and collector currents giving a large current gain (as with the common emitter configuration) therefore, providing good current amplification with very little voltage gain.



- ❑ The common emitter configuration has a current gain approximately equal to the β value of the transistor itself.
- ❑ In the CC configuration, the load resistance is connected in series with the emitter terminal so its current is equal to that of the emitter current. **The current gain in CC is slightly higher than in the common emitter configuration.**
- ❑ As the emitter current is the combination of the collector and the base current, the load resistance in this type of transistor configuration also has both the collector current and the input current of the base flowing through it.
- ❑ This type of bipolar transistor configuration is a non-inverting circuit in that the signal voltages of V_{in} and V_{out} are “in-phase”.
- ❑ The CC configuration has a voltage gain of about “1”. Thus it can be considered as a voltage-buffer.
- ❑ **Current Gain Formula (γ):** It is defined as the ratio of the emitter current (I_E) to the base current (I_B). $\gamma = \frac{I_E}{I_B}$

Relationship with β : $\gamma = 1 + \beta$ [Since $\beta = \frac{I_C}{I_B}$ and $I_E = I_B + I_C$]

Relation to Common Base Gain (α): $\gamma = \frac{1}{1 - \alpha}$; The current gain is **very high**, typically ranging from 20 to several hundreds, depending on the transistor's β value

Generalized characteristics of the different transistor configurations

Sr. No.	Parameter	CB	CE	CC
1	Common terminal between input and output	Base	Emitter	Collector
2	Input current	I_e	I_β	I_β
3	Output current	I_c	I_c	I_E
4	Current gain	$\alpha_{dc} = I_c / I_E$	$\beta_{dc} = I_c / I_B$	$\gamma_{dc} = I_E / I_B$
5	Input voltage	V_{EB}	V_{BE}	V_{BC}
6	Output voltage	V_{CB}	V_{CE}	V_{BC}
7	Voltage gain	Medium	Medium	Less than 1
8	Input resistance	Very low (20 Ω)	Low (1 $k\Omega$)	High (500 $k\Omega$)
9	Output resistance	Very high (1 $M\Omega$)	High (40 $k\Omega$)	Low (50 Ω)
10	Applications	As preamplifier	Audio amplifier	For impedance matching

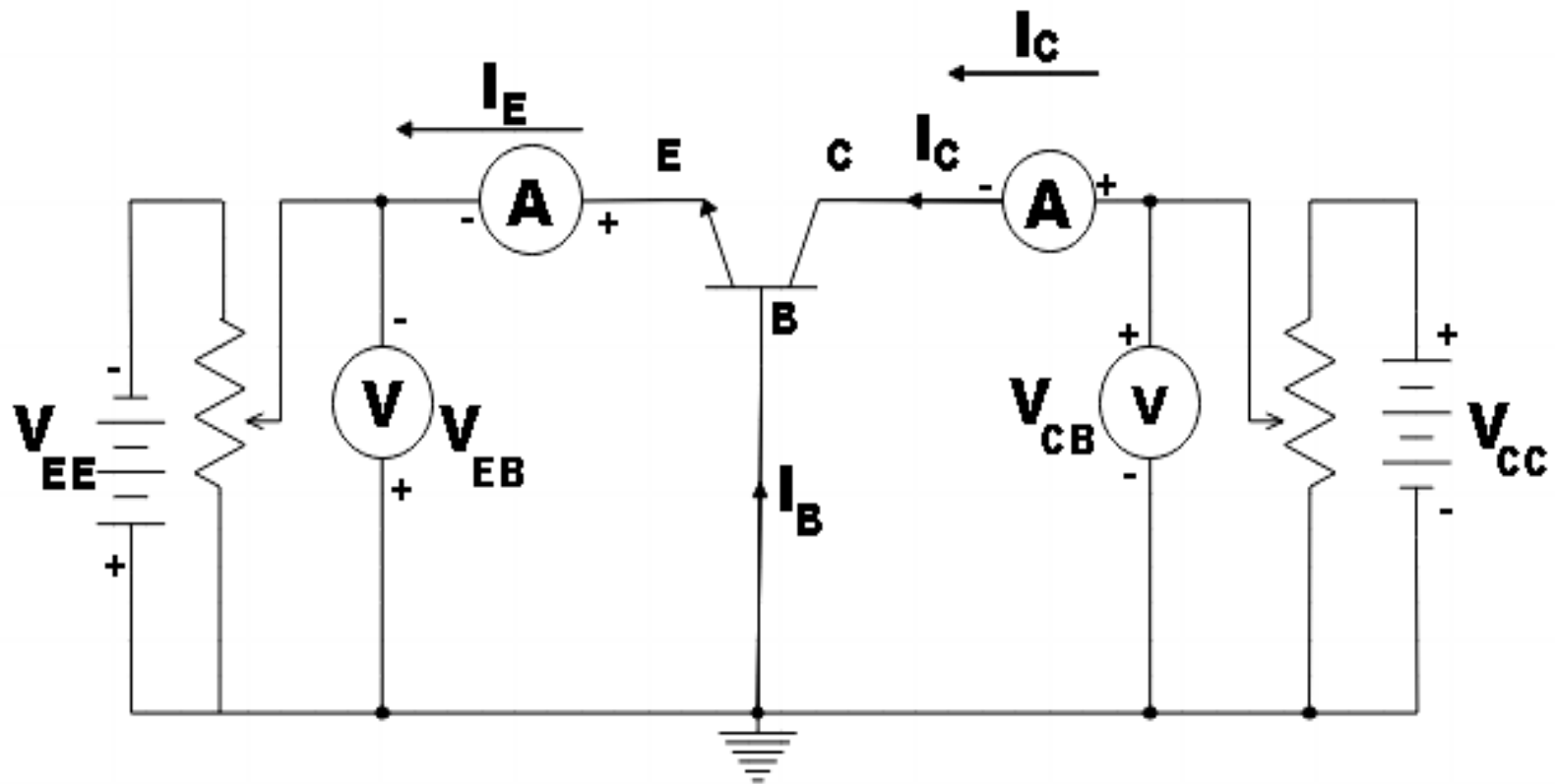
Characteristics of transistors

- There are three types of transistors characteristics
 - **Input Characteristics** (Change in Input current due to change in input voltage at fixed output voltage)
 - **Output Characteristics** (Change in output current due to change in output voltage at fixed input current/voltage)

Transfer characteristics (The relationship between a device's input and output, usually as a graph plotting an output signal (voltage or current) against an input signal (voltage or current))

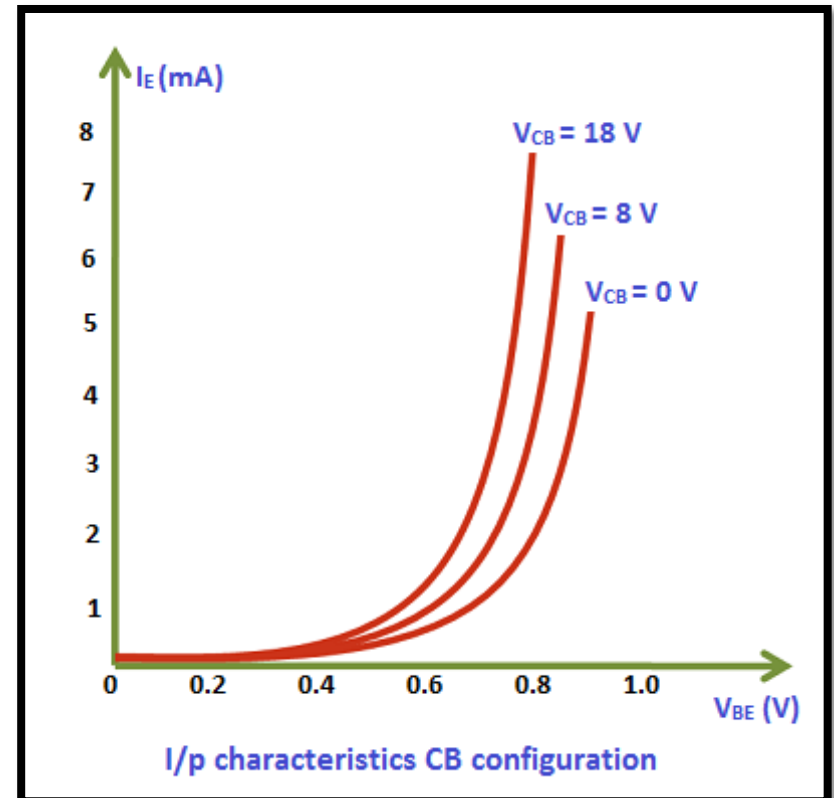
CB and CE configuration are most commonly used and are important.

CB configuration I/P and O/P characteristics



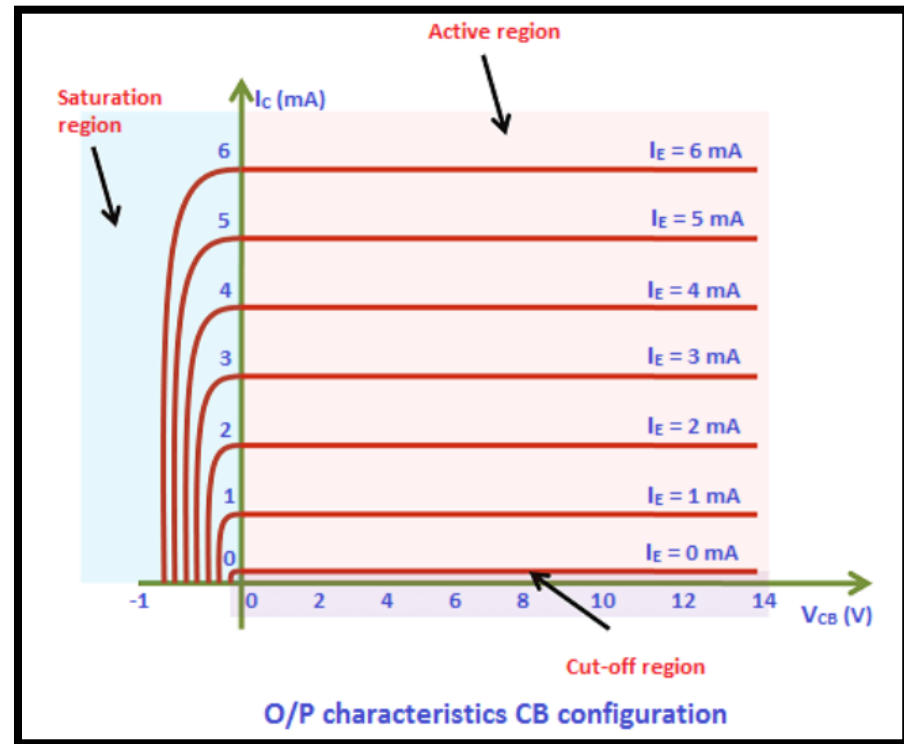
CB I/p Characteristics

- ❑ It is defined as the characteristic curve drawn between input voltage to input current whereas output voltage is constant.
- ❑ To determine input characteristics, the collector base voltage V_{CB} is kept constant at zero and emitter current I_E is increased from zero by increasing V_{EB} .
- ❑ A curve is drawn between emitter current and emitter base voltage at constant collector base voltage.
- ❑ When V_{CB} is zero EB junctions is forward biased. So it behaves as a diode so that emitter current increases rapidly and **It resembles a forward-biased diode curve.**
- ❑ Input resistance R_{in} is low (typically 50–150 ohms).



CB O/p Characteristics

- ❑ The characteristic curve drawn between output voltage to output current keeping input current constant.
- ❑ The emitter current I_E is kept constant at zero and collector current I_C is increased from zero by increasing V_{CB} .
- ❑ For a constant value of I_E , I_C is independent of V_{CB} and the curves are parallel to the axis of V_{CB} .



<i>Input Bias</i>	<i>Output Bias</i>	<i>Region of operation</i>	<i>Application</i>
Reverse	Reverse	Cut off	OFF Switch
Forward	Reverse	Active	Amplifier
Forward	Forward	Saturation	ON Switch
Reverse	Forward	Inverse Active	Attenuator

Transistor parameters

1. Input Resistance: $r_i = \frac{\Delta V_{EB}}{\Delta I_E} / V_{CB} \text{ Constant}$

$$h_{ib} = \frac{\Delta V_{EB}}{\Delta I_E} / V_{CB} \text{ Constant}$$

Input resistance in Common Base is in the order of few ohms.

2. Output Resistance: $r_o = \frac{\Delta V_{CB}}{\Delta I_C} / I_E \text{ Constant}$

$$h_{ob} = \frac{\Delta I_C}{\Delta V_{CB}} / I_E \text{ Constant}$$

3. Current Gain: $A_I = \frac{\Delta I_C}{\Delta I_E} / V_{CB} \text{ Constant}$

$$\alpha = h_{fb} = \frac{\Delta I_C}{\Delta I_E} / V_{CB} \text{ Constant}$$

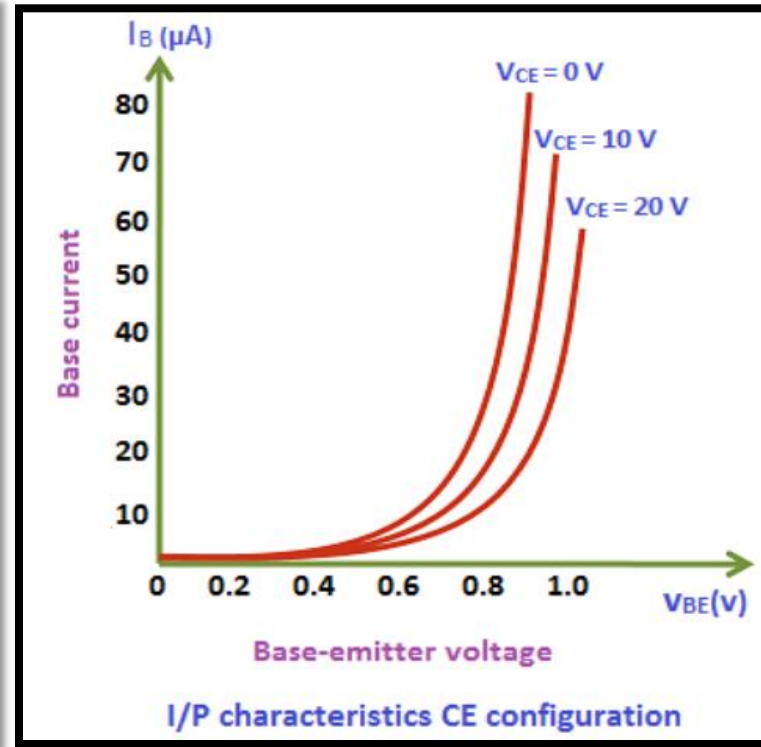
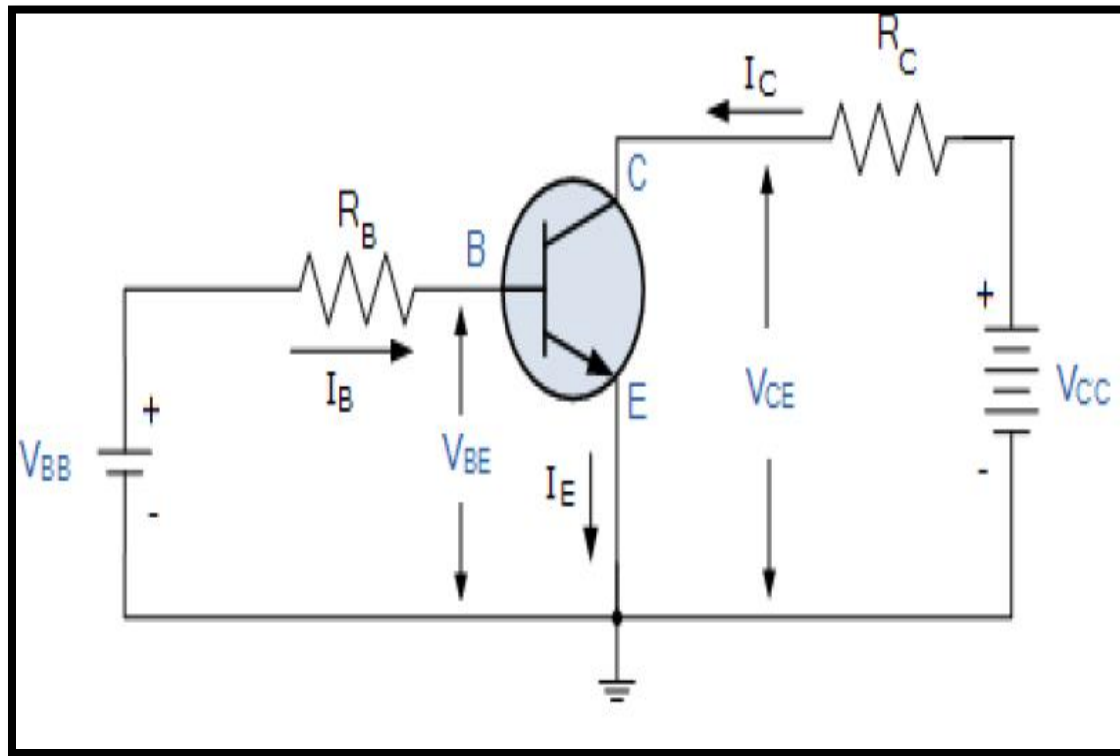
Current gain is always less than unity.

4. Voltage Gain: $A_V = \frac{\Delta V_{CB}}{\Delta V_{EB}} / I_E \text{ Constant}$

$$h_{rb} = \frac{\Delta V_{EB}}{\Delta V_{CB}} / I_E \text{ Constant}$$

Voltage gain in common base configuration is high.

Common Emitter I/p characteristics



To determine the input characteristics, the output voltage V_{CE} is kept constant at zero volts and the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded.

I/p Characteristics

❖ **Diode-Like Curve:** The input characteristic curve resembles that of a forward-biased PN junction diode. This is because the base-emitter junction is forward-biased during normal operation.

❖ **Knee (Threshold) Voltage:** There is negligible base current until the base-emitter voltage (V_{BE}) reaches a specific threshold, typically 0.7V for silicon and 0.3V for germanium transistors.

❖ **Exponential Current Increase:** Once V_{BE} exceeds the knee voltage, the base current (I_B) increases exponentially with relatively small increases in V_{BE} .

❖ **Moderate Input Resistance:** The dynamic input resistance (R_{in}) is defined as the ratio of change in V_{BE} to the change in I_B at constant V_{CE}

❖ **Value:** It is typically around 1 k Ω .

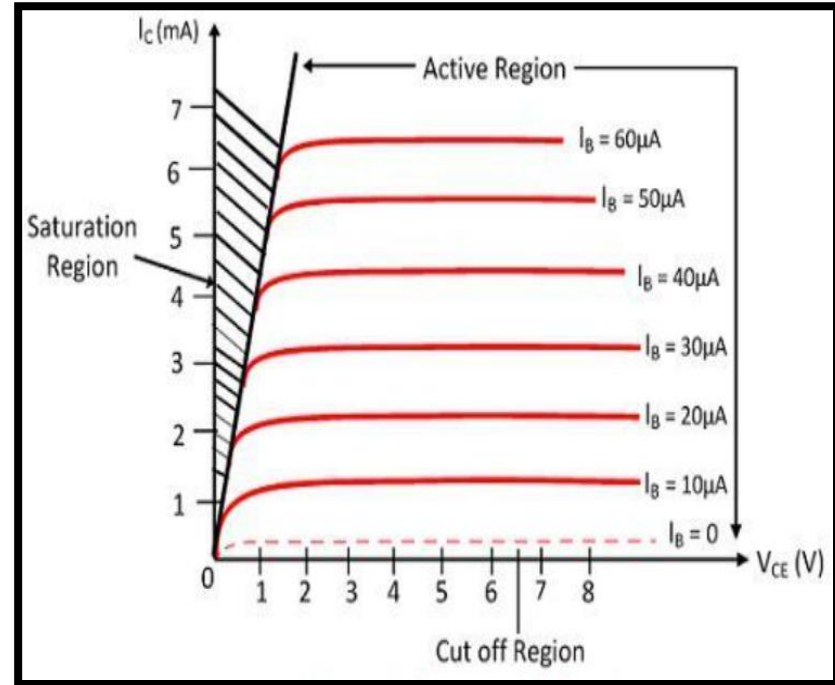
❖ **Comparison:** This resistance is higher than that of the common-base configuration but lower than the common-collector configuration.

CE configuration O/P characteristics

Three distinct operational regions:

- **Active Region:**

- **Condition:** Base-Emitter junction is forward-biased and Collector-Base junction is reverse-biased.
- **Behavior:** The curves are nearly horizontal, meaning I_C is relatively constant and mainly **depends on I_B rather than V_{CE}**
- **Application:** The transistor functions as an **amplifier** in this region.



- **Saturation Region:**

- **Condition:** Both the Base-Emitter and Collector-Base junctions are forward-biased.
- **Behavior:** I_C increases rapidly with a small increase in V_{CE} until it reaches a saturation point.
- **Application:** The transistor acts as a **closed switch**.

- **Cut-off Region:**
 - **Condition:** Both junctions are reverse-biased (I_B).
 - **Behavior:** Only a very small leakage current (I_{CE}) flows, and I_C is nearly zero.
 - **Application:** The transistor acts as an **open switch**.
- **Key Parameters & Features**
- **Output Resistance (R_o):** This is the ratio of change in V_{CE} to the change in I_C at a constant I_B . In the active region, the output resistance is **high** (typically several kilo-ohms).
- **Effect of V_{CE} (Early Effect)**
 - As the collector-emitter voltage (V_{CE}) increases, the input characteristic curves shift slightly to the right. Increasing V_{CE} expands the collector-base depletion region, effectively narrowing the base region that causes a slight decrease in base current for a given V_{BE} at higher V_{CE} levels and hence a slight slope in the curve is seen in the active region.
- **Phase Inversion:** This configuration is unique for producing a **180° phase shift** between the input and output signals.
- **Gain Characteristics:** It provides high power gain because it produces significant current and voltage gains simultaneously.

Transistor parameters

1. Input Resistance: $r_i = \frac{\Delta V_{BE}}{\Delta I_B} / V_{CE} \text{ Constant}$

$$h_{ie} = \frac{\Delta V_{BE}}{\Delta I_B} / V_{CE} \text{ Constant}$$

Input resistance in Common Emitter is moderate.

2. Output Resistance: $r_o = \frac{\Delta V_{CE}}{\Delta I_C} / I_B \text{ Constant}$

$$h_{oe} = \frac{\Delta I_C}{\Delta V_{CE}} / I_B \text{ Constant}$$

Output resistance in Common Emitter is moderate.

3. Current Gain: $A_I = \frac{\Delta I_C}{\Delta I_B} / V_{CE} \text{ Constant}$

$$\beta = h_{fe} = \frac{\Delta I_C}{\Delta I_B} / V_{CE} \text{ Constant}$$

Current gain in Common Emitter configuration is high.

4. Voltage Gain: $A_V = \frac{\Delta V_{CE}}{\Delta V_{BE}} / I_B \text{ Constant}$

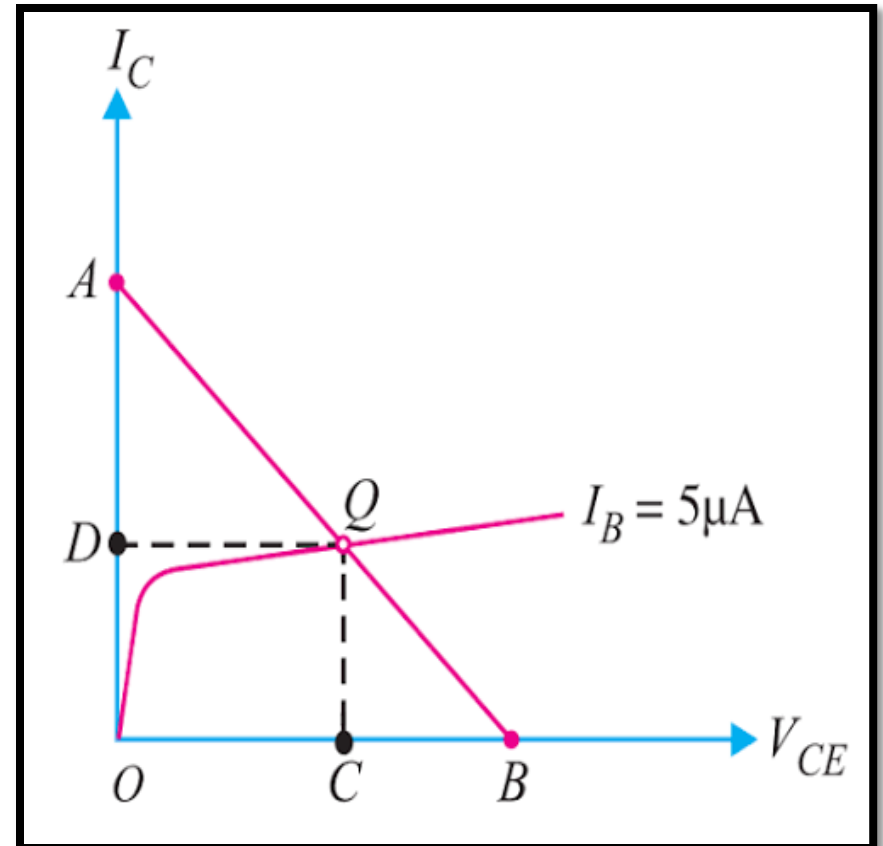
$$h_{re} = \frac{\Delta V_{BE}}{\Delta V_{CE}} / I_B \text{ Constant}$$

Voltage gain in common Emitter configuration is high.

Q Point

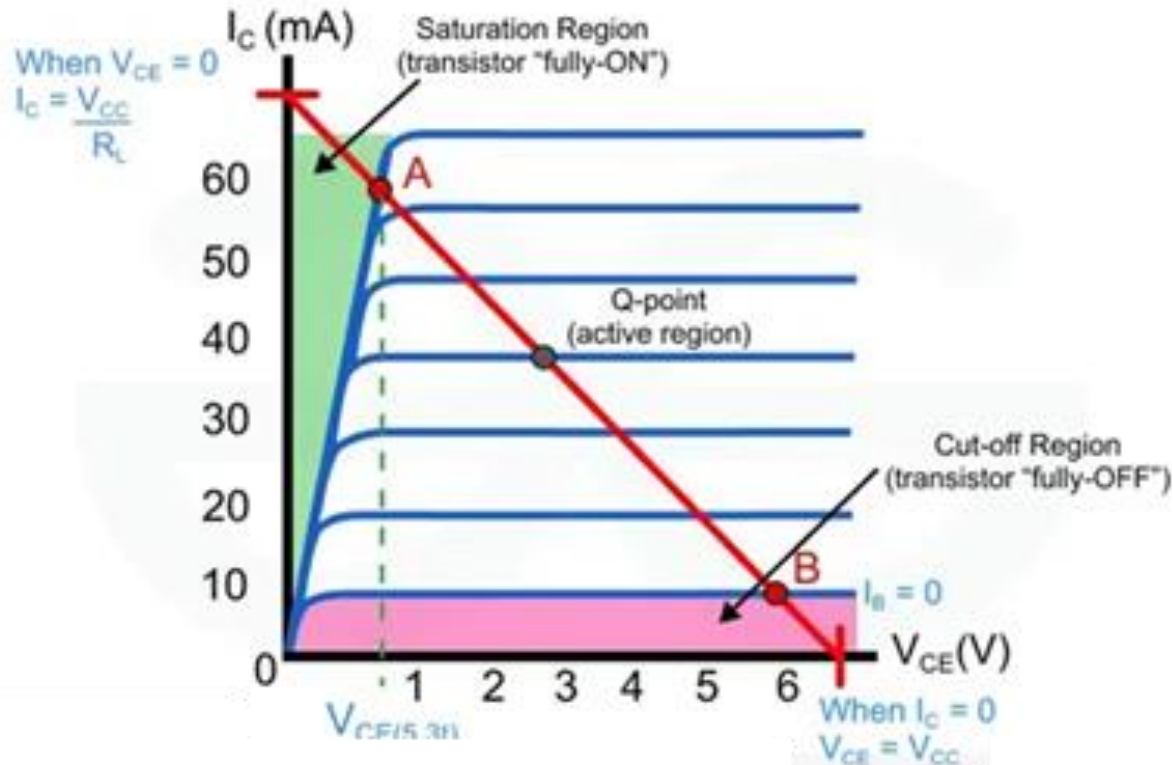
- The *quiescent point (Q-point)* is the steady DC voltage or current of a device like a transistor when no input signal is applied. It shows the normal operating condition of the device.
- The term *bias point* is used because applying a steady DC input to a transistor allows it to operate properly without cutoff or saturation while amplifying a time-varying AC signal.
- For example, a transistor could be biased to set its Q-point output to, say, 5 V to sit halfway between 0 and 10 V supply rails.
- An AC input then superimposed onto the DC level at the transistor input will generate a corresponding, amplified waveform at the output. Biasing provides room for this waveform either side of the Q-point without clipping or saturation.

- The position of the Q-point depends on the applications of the transistor.
- If the transistor is used as a switch, then for open switch the Q-point is in the cut-off region
- For the close switch, the Q-point is in the saturation region. The Q-point lies in the middle of the line for the transistor which operates as an amplifier.
- In saturation region, both the collector base region and the emitter-base region are in forward biased and heavy current flow through the junction.
- The region in which both the junctions of the transistor are in reversed biased is called the cut-off region.



Transistor as a Switch

Transistor as a switch is the simplest application of the device. A transistor can be used for switching operations or opening or closing of the circuit.

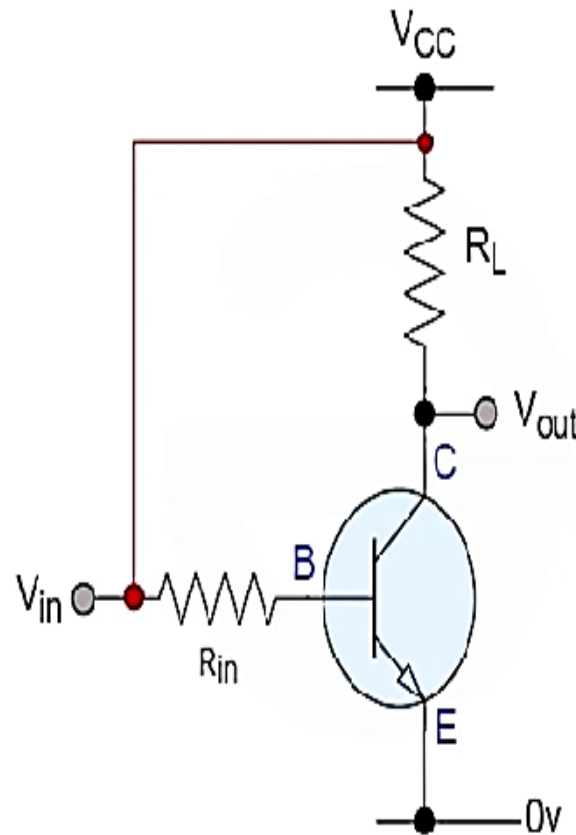


Closed switch

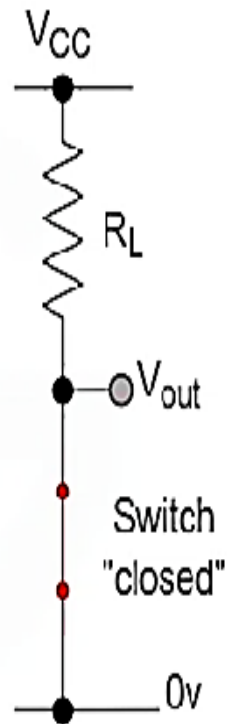
- **Saturation Region**

In this region, the transistor will be biased, so the maximum amount of the current is applied, which results in the maximum collector current results in the minimum collector emitter voltage drop which cause in depletion layer as a small as possible and maximum current flowing through the transistor, so the transistor operates in "Fully ON" switched mode.

- The base and input are connected to V_{CC} .
- Base-Emitter voltage $V_{BE} > 0.7V$.
- Base-Emitter junction is forward biased.
- Base-collector junction is forward biased.
- Transistor is "fully-On"



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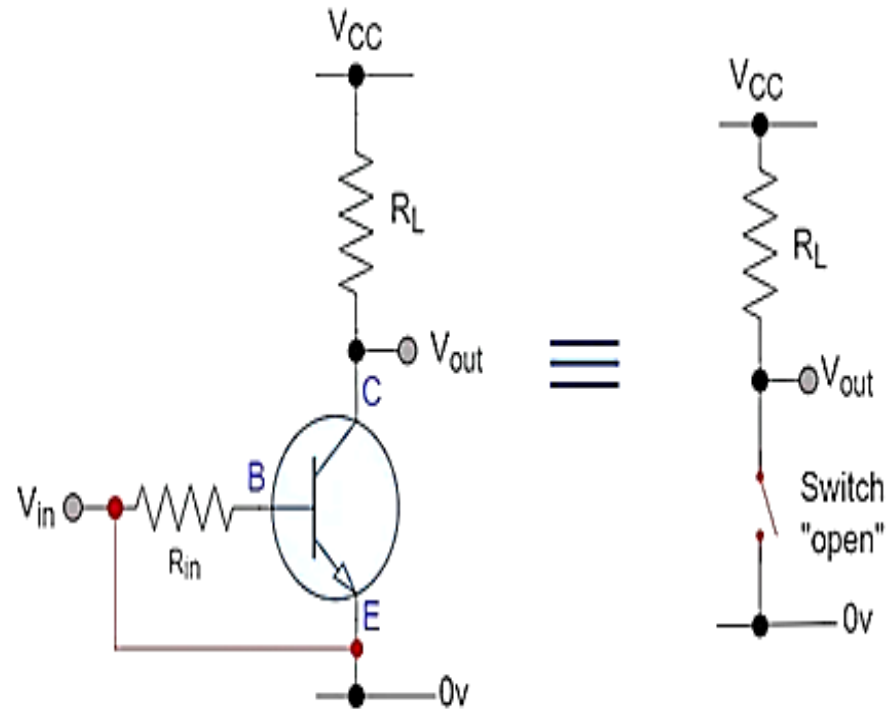


- Max Collector current flows ($I_c = V_{CC}/R_L$)
- $V_{ce} = 0$ (ideal saturation)
- $V_{out} = V_{ce} = "0"$.
- Transistor operates as a closed switch.

Open switch

- **Cut-off Region**
- In the cut-off region of the transistor, it operates under conditions of zero input base current (I_B), zero output collector current (I_C), and maximum collector voltage (V_{CE}). This configuration creates a significant depletion layer, preventing the flow of current through the device and effectively switching it off.

- The input and base are grounded.
- Base emitter voltage $V_{be} < 0.7V$
- Base Emitter junctions is reversed.
- Base- collector junction is reversed biased.
- Transistor is "fully-Off" (cut off region)
- No collector current flows ($I_c = 0$)



- $V_{out} = V_{ce} = V_{cc} = 1$
- Transistor operates as open switch. In the cut off region of off mode when using a bipolar transistor as a switch